

## PERM BOLSHOYE SAVINO, Russian Federation

WMO: 282250

Lat: 57.915N

Lon: 56.021E

Elev: 404

StdP: 14.48

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -23.4      | -16.9 | -29.2                         | 1.1 | -23.0 | -23.4 | 1.5 | -17.2 | 25.2                  | 18.6 | 23.2 | 18.8 | 4.5                   | 240  | 0.666 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.3                   | 86.0            | 68.6 | 82.5 | 66.7 | 79.0 | 65.0 | 70.4                | 81.6 | 68.8 | 79.7 | 66.9 | 76.7 | 7.8                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.5                            | 98.9 | 75.7 | 64.7 | 92.7 | 73.3 | 62.9 | 87.0 | 71.4 | 34.7          | 81.8 | 33.2 | 79.5 | 31.8 | 76.1 | 76.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |  |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|--|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |  |
| Min               |      | Max  |                            | Min   | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |  |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |
| 22.5              | 20.1 | 17.9 | DB                         | -29.1 | 90.1 | 6.9                | 3.6                                                | -34.0     | 92.7 | -38.0      | 94.8 | -41.9      | 96.8 | -46.9      | 99.3 |  |
|                   |      |      | WB                         | -29.8 | 73.0 | 6.7                | 2.4                                                | -34.6     | 74.7 | -38.5      | 76.1 | -42.3      | 77.5 | -47.2      | 79.2 |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 37.5   | 8.7   | 9.9   | 24.5  | 39.0  | 53.0  | 61.2  | 65.5  | 61.0  | 50.9  | 38.1  | 23.0  | 12.7  |
|                                                                               | DBStd       | 22.62  | 13.85 | 13.37 | 8.43  | 9.16  | 9.20  | 8.01  | 6.97  | 7.62  | 7.64  | 8.46  | 12.28 | 13.83 |
|                                                                               | HDD50       | 6057   | 1279  | 1124  | 791   | 346   | 78    | 9     | 1     | 4     | 80    | 380   | 809   | 1156  |
|                                                                               | HDD65       | 10262  | 1744  | 1544  | 1256  | 779   | 382   | 165   | 81    | 173   | 425   | 833   | 1259  | 1621  |
|                                                                               | CDD50       | 1476   | 0     | 0     | 0     | 17    | 171   | 344   | 482   | 344   | 107   | 11    | 0     | 0     |
|                                                                               | CDD65       | 209    | 0     | 0     | 0     | 0     | 11    | 50    | 98    | 48    | 2     | 0     | 0     | 0     |
|                                                                               | CDH74       | 2147   | 0     | 0     | 0     | 4     | 170   | 489   | 1006  | 454   | 24    | 0     | 0     | 0     |
|                                                                               | CDH80       | 601    | 0     | 0     | 0     | 0     | 28    | 125   | 302   | 144   | 2     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 8.9    | 9.5   | 9.0   | 10.1  | 9.4   | 9.0   | 8.2   | 6.8   | 7.4   | 8.2   | 9.6   | 9.5   | 9.5   |
| Precipitation                                                                 | PrecAvg     | 31.5   | 2.4   | 1.7   | 2.0   | 1.7   | 2.4   | 3.4   | 3.2   | 3.3   | 2.8   | 3.1   | 2.9   | 2.5   |
|                                                                               | PrecMax     | 40.9   | 4.1   | 3.3   | 3.5   | 3.0   | 4.2   | 6.3   | 6.4   | 7.2   | 5.2   | 5.7   | 5.1   | 4.2   |
|                                                                               | PrecMin     | 24.9   | 0.6   | 0.2   | 0.2   | 0.0   | 0.9   | 1.5   | 0.7   | 0.9   | 0.6   | 1.4   | 0.7   | 1.0   |
|                                                                               | PrecStd     | 3.6    | 0.8   | 0.8   | 0.9   | 0.8   | 0.9   | 1.2   | 1.4   | 1.3   | 1.1   | 1.1   | 1.0   | 0.8   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 34.2  | 35.7  | 46.1  | 73.3  | 84.0  | 88.0  | 89.8  | 89.9  | 77.2  | 64.8  | 48.6  | 36.0  |
|                                                                               |             | MCWB   | 33.7  | 33.4  | 38.6  | 55.5  | 62.7  | 69.2  | 70.0  | 68.2  | 59.9  | 52.4  | 47.0  | 35.3  |
|                                                                               | 2%          | DB     | 32.4  | 33.7  | 41.0  | 64.8  | 78.7  | 83.3  | 86.3  | 84.0  | 71.6  | 58.9  | 44.2  | 34.1  |
|                                                                               |             | MCWB   | 31.7  | 32.7  | 35.5  | 51.3  | 59.4  | 67.2  | 69.2  | 67.1  | 59.7  | 50.7  | 42.6  | 33.5  |
|                                                                               | 5%          | DB     | 30.4  | 31.7  | 37.7  | 59.1  | 74.8  | 80.2  | 83.8  | 78.8  | 67.6  | 53.6  | 39.3  | 33.2  |
|                                                                               |             | MCWB   | 29.5  | 30.5  | 34.1  | 48.1  | 57.9  | 65.8  | 68.2  | 65.9  | 57.6  | 48.1  | 38.2  | 32.6  |
|                                                                               | 10%         | DB     | 26.8  | 28.3  | 35.6  | 53.7  | 70.0  | 75.4  | 80.3  | 74.1  | 62.7  | 49.9  | 35.8  | 30.0  |
|                                                                               |             | MCWB   | 26.0  | 26.9  | 33.0  | 44.4  | 55.6  | 63.1  | 66.8  | 64.1  | 55.7  | 46.8  | 34.6  | 29.1  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 34.0  | 33.9  | 39.4  | 56.4  | 65.9  | 72.2  | 73.1  | 71.1  | 63.5  | 54.8  | 47.4  | 35.7  |
|                                                                               |             | MCDB   | 34.1  | 34.3  | 44.2  | 69.4  | 78.3  | 83.9  | 84.3  | 82.4  | 71.0  | 61.3  | 48.5  | 35.8  |
|                                                                               | 2%          | WB     | 32.0  | 32.3  | 36.5  | 52.3  | 62.1  | 69.3  | 71.1  | 69.1  | 60.8  | 51.4  | 41.9  | 33.7  |
|                                                                               |             | MCDB   | 32.4  | 33.2  | 39.0  | 62.8  | 74.3  | 80.4  | 82.2  | 80.2  | 68.8  | 57.2  | 43.1  | 34.0  |
|                                                                               | 5%          | WB     | 29.3  | 30.0  | 34.7  | 49.0  | 59.5  | 66.9  | 69.5  | 67.0  | 58.8  | 49.0  | 37.9  | 31.9  |
|                                                                               |             | MCDB   | 30.1  | 31.3  | 36.8  | 58.0  | 71.3  | 76.8  | 80.3  | 76.5  | 65.8  | 52.7  | 38.7  | 32.5  |
|                                                                               | 10%         | WB     | 26.0  | 26.6  | 33.1  | 45.4  | 57.1  | 64.7  | 67.7  | 64.8  | 56.6  | 46.6  | 34.6  | 28.7  |
|                                                                               |             | MCDB   | 26.8  | 27.8  | 35.1  | 52.5  | 67.4  | 73.5  | 78.2  | 72.9  | 61.9  | 49.8  | 35.3  | 29.4  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 12.0  | 13.9  | 14.4  | 17.5  | 20.8  | 18.8  | 19.3  | 17.4  | 14.8  | 9.8   | 8.9   | 10.6  |
|                                                                               |             | MCDBR  | 10.4  | 9.9   | 14.7  | 26.2  | 27.7  | 24.0  | 25.4  | 24.4  | 21.7  | 17.2  | 8.8   | 6.9   |
|                                                                               |             | MCWBR  | 10.4  | 9.1   | 10.8  | 14.4  | 13.1  | 11.3  | 11.2  | 11.1  | 11.8  | 10.6  | 8.1   | 7.1   |
|                                                                               | 5% WB       | MCDBR  | 10.5  | 9.8   | 12.1  | 24.3  | 23.8  | 21.8  | 22.6  | 22.1  | 18.8  | 15.0  | 8.7   | 7.2   |
|                                                                               |             | MCWBR  | 10.7  | 9.3   | 9.6   | 14.1  | 13.0  | 11.5  | 10.9  | 11.0  | 11.1  | 10.2  | 8.3   | 7.6   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.293  | 0.320 | 0.375 | 0.405 | 0.395 | 0.391 | 0.414 | 0.428 | 0.380 | 0.350 | 0.311 | 0.289 |       |
|                                                                               | taud        | 1.995  | 2.001 | 1.926 | 2.071 | 2.292 | 2.359 | 2.303 | 2.234 | 2.332 | 2.266 | 2.095 | 1.970 |       |
|                                                                               | Ebn at Noon | 159    | 214   | 233   | 252   | 266   | 269   | 258   | 244   | 239   | 208   | 153   | 119   |       |
|                                                                               | Edh at Noon | 19     | 30    | 43    | 44    | 38    | 36    | 37    | 37    | 29    | 23    | 17    | 14    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 168    | 447   | 858   | 1270  | 1617  | 1721  | 1729  | 1290  | 761   | 326   | 138   | 91    |       |
|                                                                               | RadStd      | 26     | 73    | 105   | 169   | 180   | 153   | 183   | 128   | 80    | 47    | 27    | 19    |       |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (10 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page